

An open source FORTRAN90 Code Using Vortex Rings and a Vortex Cylinder For A Constantly Loaded Actuator Disk *

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Abstract

An open source FORTRAN90 implementation of a vortex ring/-cylinder code applied to a constantly loaded actuator disk is described. Results for $c_T = 1$ (propeller) and $c_T = -8/9$ (Betz wind turbine) are presented and compared with data from Prof. em. Gijs van Kuik [9]. It is available via github:

<https://github.com/Schaffarczyk/RingsCylinderActuatorDisk/>

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1 Introduction

Rotors and especially those of wind turbines are modeled by an axi-symmetric actuator disk for more than 150 years, see for example [19, 14]. Originally, only the axial velocity component was used in an integral momentum theory description [15] but it easily can be shown that this not consistent with the differential equations for fluid flow, in case of neglecting any influence of viscosity, Euler's Equation:

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{f}. \quad (2)$$

Therefore, to include the radial velocity component vortex rings methods [3, 17] seem to be the simplest way to overcome this shortcoming if one does not want to solve the differential equations itself numerically.

Beginning with Stig Øye's paper from 1990 [11] this approach has been further developed and applied by [18, 1, 2]. It was generalized to non-uniform loading by Conway [8].

Here we report on an independent FORTRAN 90/95 [6] implementation of this model.

Main objectives are:

- scrutinize the accuracy of the model,
- try to find hints of the properties of a possible edge singularity [16, 7].

2 Implementation

2.1 Model description

We use 2D real axi-symmetric coordinates z, r with velocity $\mathbf{u} = (u_z, u_r)$. The Actuator Disk is located at $z = 0$ and $0 \leq r < 1$. The edge ($\mathbf{r_E} = (0, 1)$) may carry a singularity, a point where $\mathbf{u}$ is not defined.

The model itself is defined by a piece-wise linear vortex sheet which starts at $\mathbf{r_{edge}} = (0, 1)$ and ends at $\mathbf{r_{Cyl}} = (zsivc, rsivc)$, the location where a semi-infinite vortex-cylinder is appended. For $0 < z < zsivc$ NP vortex rings with strength γ are placed with non-uniform distance at locations zsl, rsl . This

part of our approach is identical with that from [1, 2]. The locations of the z-positions of the rings are distributed according to

$$z(i) = z_{SIVC} * (1. - \cos(\varphi)) \quad (3)$$

$$\varphi = 0.5 \cdot \pi \cdot i / np \quad (4)$$

$$1 \leq i \leq np \quad (5)$$

To circumvent singularities on the slipstream (i.e. infinite velocities) all velocities are evaluated just in the (arithmetic) middle of two slip-stream (singularity carrying) points. This approach circumvents introducing an artificial *vortex core*.

Basic model parameter is the thrust coefficient c_T where $c_T > 0$ refers to the propeller case (induced velocities > 0) and $c_T < 0$ to the turbine case. Momentum theory is valid for $-1 \leq c_T < \infty$. Many parameters can be deduced from momentum theory, for example the asymptotic radial extension of the slip-stream, which is given by

$$r_{z \rightarrow \infty}^{sl} \equiv RSIVC = \sqrt{\frac{1 + \sqrt{1 + c_T}}{2 \cdot \sqrt{1 + c_T}}} \quad (6)$$

$$\text{For } c_T = 1 \text{ we have: } r_\infty = 0.9239 \text{ and} \quad (7)$$

$$\text{for } c_T = -8/9 : r_\infty = \sqrt{2} = 1.4142 . \quad (8)$$

Value for the stream function at infinity:

$$\psi_{wake} = \frac{c_T}{4 \cdot \sqrt{1 + c_T}} \quad (9)$$

Our model defining parameters are summarized in tab 1. Inflow velocity is set to $u(r)_{z \rightarrow -\infty} = 1.0$.

2.2 Induced Velocities

Accurate calculation of induced axial and radial velocities is the key for any arbitrary accurate (= successful) model. Calculation is divided into three parts:

1. Induced velocities from the NP-1 other vortex rings to the regarded one for $0 \leq z \leq ZSIVC$,

name	meaning
major	
c_T	thrust coefficient
restart	LOGICAL. if .T. code will restart from existent sls.DAT
NP	number of rings
ZSIVC	axial location of semi-infinite cylinder
minor	
selfind	(0,1), switch on/off self-induction
update	(0,1), def = 1, debug only
maxiter	maximum number (< 100) of slip-stream iterations
under	under relaxation (0.1...1.)
epscP	
wakety	initial wake type (3 models available)
wakety=1	$rsl(i) = rinf + (1.-rinf)*exp(-al*zsl(i))$
wakety=2	gnuplot fit of vK data $cT = 1$ and $cT = -8./9$. only
wakety=3	van Kuik's Eq. (D.10)
wake-a	parameter al for wake-ty = 1
iter-sc	slip-stream iteration (3 = psi)

Table 1: Table of major and minor influencing parameters

2. Self-induced velocities of a regarded ring:

We only have implemented Bontempo's [2] Eqs. (6) and (7) which surprisingly do not contain any core-parameter:

$$v_{z,m \rightarrow m} = \gamma_m \left\{ -\frac{\beta_{m+1} - \beta_m}{8\pi} \cos \beta_m - \frac{ds_m}{4\pi r_m} \left[\log \left(\frac{8\pi r_m}{ds_n} \right) - 0.25 \right] \right\} \quad (10)$$

$$v_{r,m \rightarrow m} = \gamma_m \left\{ \sin(\beta_m) - \frac{\beta_{m+1} - \beta_m}{8\pi} \right\} \quad (11)$$

with β_m being the slope at ring m

with r_m being radial distance of ring m

with s_m being the distance of the two rings nearby

(12)

For more details on self-induction on vortex rings, see, for example [13].

3. Induced velocities from a semi-infinite vortex cylinder with tangential vorticity located at ZSIVC [3], chapter 36.

To avoid errors in implementation three different *models* (i.e. set of equations) can be chosen:

1. Eqs. (1) to (7) from [1, 2] called model *Bontempo*,
2. Eqs. (D1) from [17] called model *van Kuik* and
3. Eqs. from chapters (33) and (36) from [3] called model *Branlard*.

To be as consistent in nomenclature as far as possible we throughout prefer (and use) equations from Branlard [3].

2.3 Slip Stream Iteration

The shape of the slipstream is determined by the induced velocities and vice versa by constraints described in [1, 2, 17]:

- force free condition implemented as $\gamma = c_t / (2 \cdot u_s)$ with $u_s^2 = (1 + u_z)^2 + u_r^2$,
- slip stream aligned with flow, implemented as $\psi_w(z) = \psi_w(\infty)$.

An initial slip-stream shape may be helpful to reduce numbers of iterations. The following models are implemented

1. A cylinder with $RSIVC = 1$ or
2. a prescribed exponential shape described by

$$r_{sl}(z) = RSIVC + (r_0 - RSIVC \cdot \exp(-a \cdot z)) \quad . \quad (13)$$

3. A prescribed algebraic shape deduced from the axial velocity at $r=0$ for a vortex cylinder (attributed by van Kuik [17] to Snel and Schepers, which probably

$$\text{or from } u_z(r=0) \frac{\gamma}{2} \left(1 + \frac{z}{\sqrt{(z^2 + RSIVC^2)}} \right) \quad (14)$$

$RSIVC$ is given by Eq. (8) but a (not to be confused with axial induction) has to be guessed $\approx 1...2$. Fig. 1 shows a GNUPLOT fit from van Kuik's $cT = 1$ data. As he sets $RSIVC_{vk} := 1$ a fit results in:

1. $r_0 = 1.03498$,
2. $RSIVC = 1.00006$ (as it should) ,
3. $a = 1.01452$.

We have implemented three different algorithms for the slip-stream iteration:

1. Eqs. (13) and (14), Fig. 2 from [2],
2. Shift points normal to actual slip-stream direction or
3. Use van Kuik's [17] Eq. (D.12).

$$\boxed{\partial r(z) = \frac{\partial \psi}{r \cdot u_z}}$$

with $\partial \psi = \psi(\text{slip} - \text{stream}, z) - \psi_{wake}$ from Eq. 9.

Method No. (3) is to be preferred All algorithms may be under-relaxed. Corresponding factors have to be guessed:

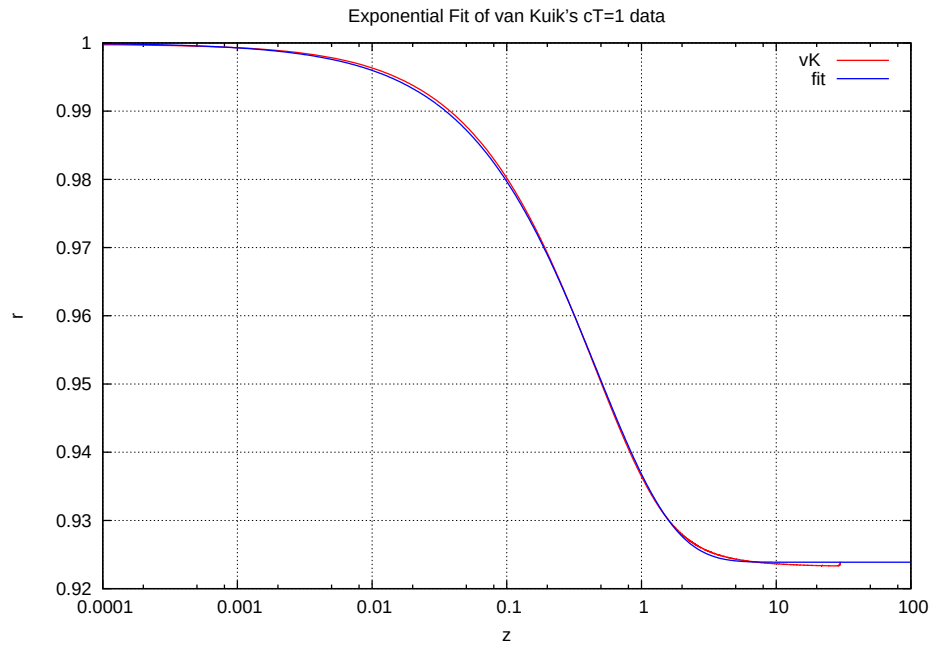


Figure 1: Two parameter a_2, k -fit of van Kuik's $c_T = 1$ slip stream according to $r(z) = r_{inf} + (1. - r_{inf}) * \exp(-a_2 * z^k)$. Here: $a_2 = 1.7743$ and $k = 0.7570$.

1. $0.001 \cdots 0.01$,
2. < 0.01 ,
3. $0.1 \cdots 0.9$.

Slip-stream iteration ends if

1. number of iterations MAXITER is exceeded or
2. $\text{errcP} := (c_P \text{ from code} - c_{P,mom}) \leq EPSCP$.

Compared to older versions (from 2021) improvement slip-stream iteration and accuracy was possible by a careful choice of under-relaxation parameter and increasing the number of vortex rings.

3 Numerics

3.1 The goal of numerical computations

Everybody knows from own experience the NEWTON-(Raphson) method for finding zeros of functions ($f(x_0) = 0$) or SIMPSON's rule for 1D-Integration. The ultimate manifestation of numerical algorithms may be found in the famous (series of) text book(s): *NUMERICAL RECIPES in Fortran 77* [12].

Let us state

Theorem 1. *A numerical schema is meaningful if and only if,*

$$\forall \epsilon > 0 \exists n \in \mathbb{N}$$

that after n iterations the result is accurate by ϵ .

In a new version we have some reasonable hope that the proposed code is able to obey this theorem.

3.2 A short note on FORTRAN [6]

We will not defend ourselves why we use the *Grandfather of all (higher) programming languages* but only use it. It has been tried to write a code as simple as possible to make understanding and changes (**i.e. maintainability**) as simple as possible. A minimum set of algorithmic features is used. Especially, no tricks from Computer Science have been used.

3.2.1 Compilation

For compilation we use (in a LINUX environment) *GNU Fortran* (in short: gfortran) [10] with mandatory options (see **compVortexCode.cmd**):

- fno-automatic helps to resolve notorious issues with local variables in SUBROUTINES and FUNCTIONS
- fdefault-real-8 set all floating numbers to REAL*8 (= 64 bit). This then should ensure an internal accuracy of 52 bits (more than 30 decimal digits) and a computational range of $\pm 10^{\pm 302}$

3.2.2 Modularization

The code is divided into 3 pieces only:

- mem.f, declares global variables,
- VortexCode.f, main slip stream iteration loop ,
- SubA.f, a set of used functions, including evaluation of Elliptic Integrals (from an external source).

3.3 Elliptic Integrals

Calculation of Elliptic Integrals, see chapter (C.4) of [3], is performed via an open-source subroutines **RFL** by [5]

```
c***begin prologue  rfl
c***date  written    790801    (yymmdd)
c***revision  date   831228    (yymmdd)
c***category    no.   c14
c***keywords    duplication theorem, elliptic integral, incomplete, complete
c               integral of the first kind, taylor series
c***author    carlson, b.c., ames laboratory-doe
c               iowa state university, ames, iowa  50011
c               notis, e.m., ames laboratory-doe
c               iowa state university, ames, iowa  50011
c               pexton, r.l., lawrence livermore national laboratory
c               livermore, california  94550
```

It was tested and compared with MATHEMATICA to 10 digits.

Restart	. false .
N-Ring	8000
ct	-0.888888
zsivc	200
selfind	. true .
update	. true .
maxiter	10
under	0.1
epscp	1.e-5
IndVel	3
wakety	3
wake-a	1.0
itersc	3

Table 2: Sample input (configuration) file

3.4 Evaluation of induced velocities

Testing the accuracy of the induced velocities was done separately in the following manner: using all cases from table 3:

- check $v_{z,r}$ for the ring part $0 < z < zsivc$
- check $v_{z,r}$ for the cylinder part $z \leq zsivc$
- compare to data from van Kuik [9].

3.5 Verification and Evaluation

Power is determined by using Bontempo's Eqs. (15) to (20) [2]. Radial integration of $u_z(z = 0, r)$ from $r = 0$ to $r = 1$ is performed by Simpson's rule. This schema is accurate to 2nd order. Thus, for an accuracy of $< 10^{-6}$ at least 10^3 evaluation points are needed. We use 2000 point.

A computational case is defined in **inpa.dat** by use of the following set-up parameters:

Before testing the iteration schema two types of a fixed shape for the slipstream can be examined

1. A fixed cylinder with $r = 1$ or $r = r_{SIVC}$

2. prescribed shape from Eq.(13), see Fig. 1 but for $c_T = 1$ and $c_T = -8/9$ only).
3. An algebraic shape according to Eq. 14.

4 Output Files, Tools for Vizualation and Testing

4.0.1 Description of Output Files

Following (ASCII) output is provided (list not complete):

1. sls.DAT: geometry, velocity, vortex strength and stream-function on vortex-sheet,
2. vzd.DAT, vrd.DAT: velocities at the disk ($z=0$),
3. vza.DAT: axial velocity at the axis ($r=0$),
4. psid.DAT: stream-function at the disk ($z=0$),
5. streaml.DAT: 3D data for contour plots of streamline (by str-cont.py).

4.0.2 Description of GNUPLOT scripts

1. SLS-vz-(Prop,WT).gpl and SLS-vr-(Prop,WT).gpl: plot shape and velocities of slip-stream
2. vza.gpl and vzd.gpl: plot axial and radial velocity at disk
3. gamma.gpl: plot vortex strength
4. and more.

Some GNUPLOT scripts (waked.gpl and)have to be executed via:

```
gnuplot -e "ct=1.0; load 'waked.gpl'"
```

to allow for input pf c_T .

4.0.3 One Python Script

1. str-cont.py: plot contour lines of stream function together with slip-stream (red).

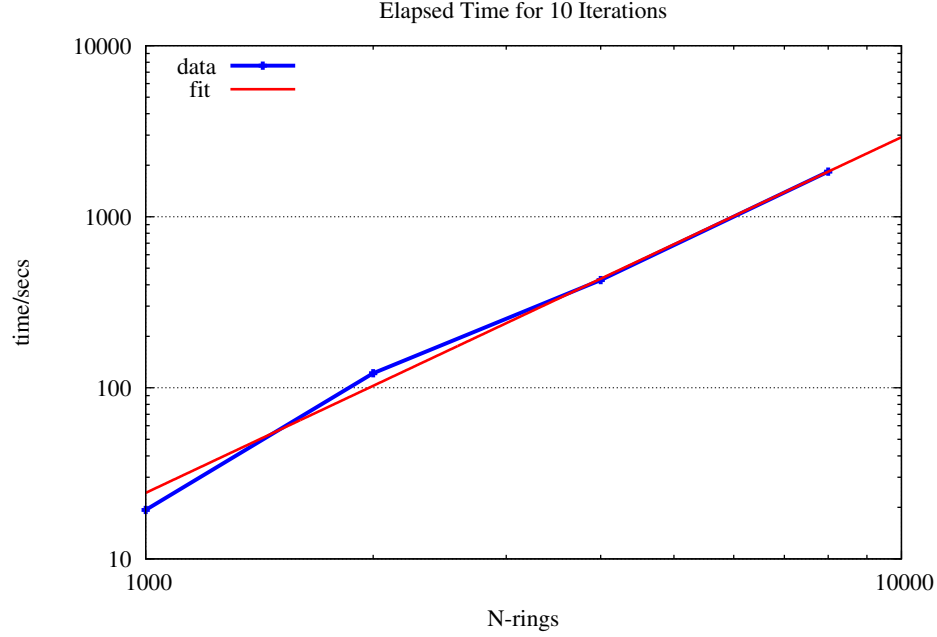


Figure 2: Elapsed CPU time for 10 slip stream iterations. Hardware: Intel i3-4160 at 3.6 GHz. Operating system: openSUSE Leap 42.3 Linux, Kernel 4.12.14-lp151.28.44-default

4.0.4 Run time estimation

As expected, CPU-time scales approximately $\sim NP^2$, see Fig. 2:

5 Results for sample cases

5.0.1 Sample case 1: Propeller with $c_T = 1$

As can be seen from the numbers above, this case is only moderately non-linear. Slip stream contraction is down to approx. 0.93 and $c_T = 1.207$, compared to a linear approximation [4], Eq. (9.107), which gives $c_{P,lin} = 1 + c_T/4 = 1.25$

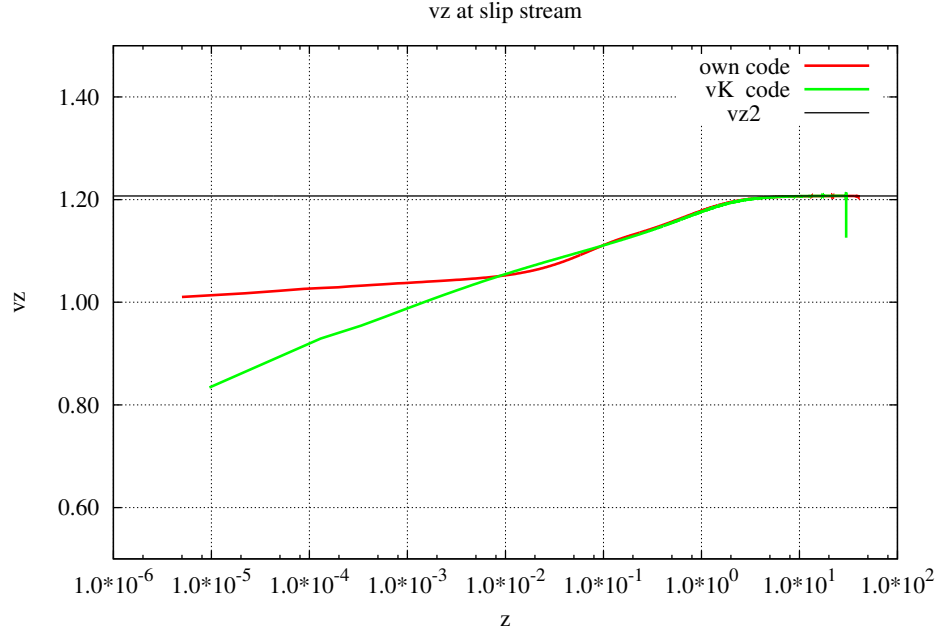


Figure 3: Calculated axial distribution of induced axial velocity, $cT = 1$.

Some sample figures: Fig. 3 shows a comparison of:

1. induced axial velocity at the slip stream, Fig. 3
2. vortex strength at the slip stream, Fig. 4
3. stream lines and slip stream, Fig. 5

with van Kuik [9] data.

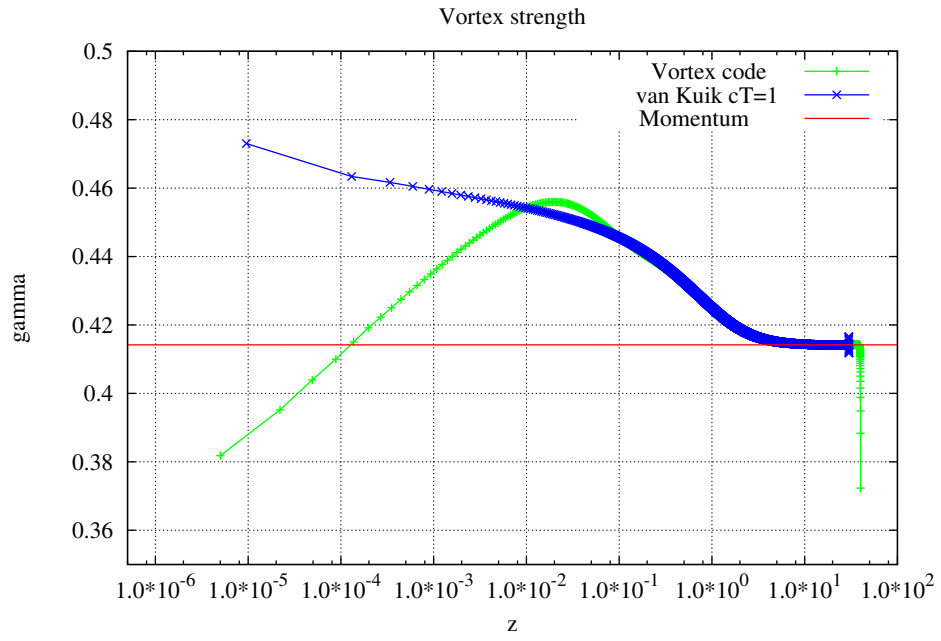


Figure 4: Calculated axial distribution of vortex strength, $cT = 1$.

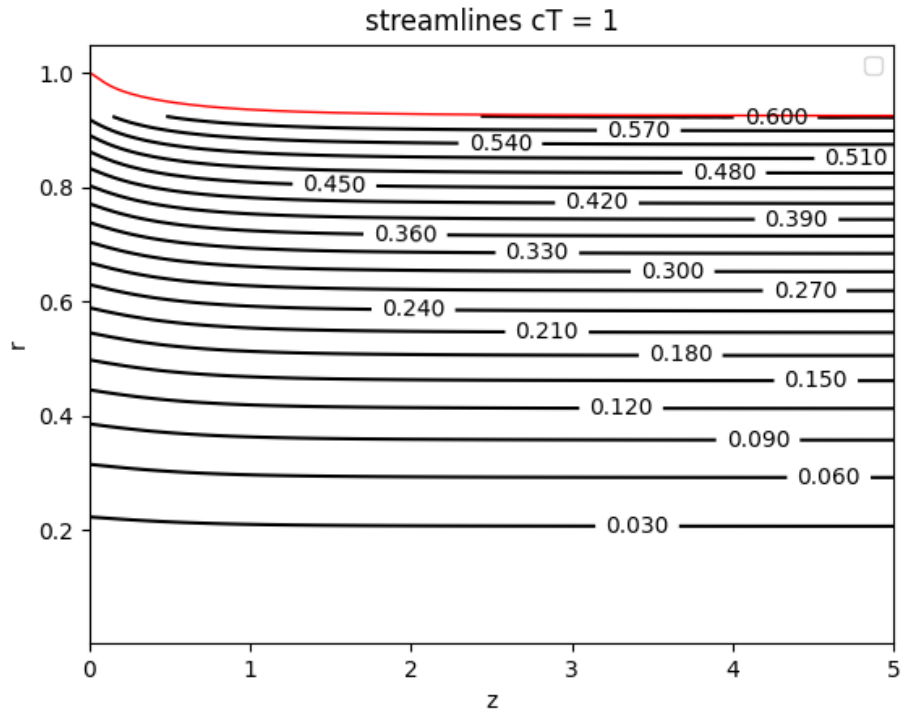


Figure 5: Calculated stream lines (lines of $\Psi = \text{const.}$) for $c_T = 1$. In red: wake boundary. As can be seen the edge streamline coincides well with $\Psi_{wake} = 0.60355339$.

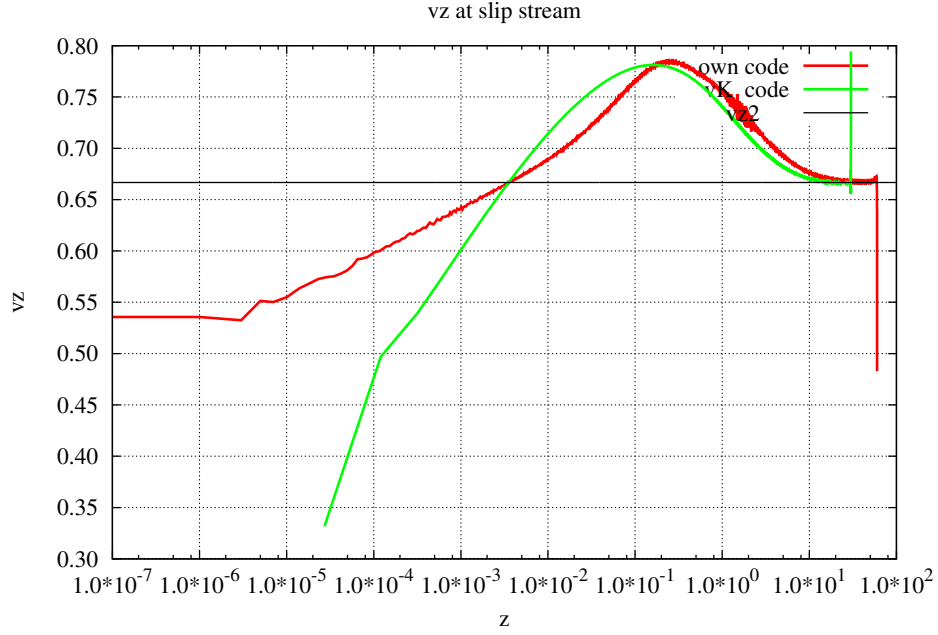


Figure 6: Calculated axial distribution of induced axial velocity, Betz case

5.0.2 Sample case 2: (Betz-)Turbine with $c_T = -8/9$

Same plots (with one addition) as shown in subsection 5.0.1:

1. induced axial velocity at the slip stream, Fig. 6
2. induced radial velocity at the slip stream, Fig. 7
3. vortex strength at the slip stream, Fig. 8
4. stream lines and slip stream, Fig. 9

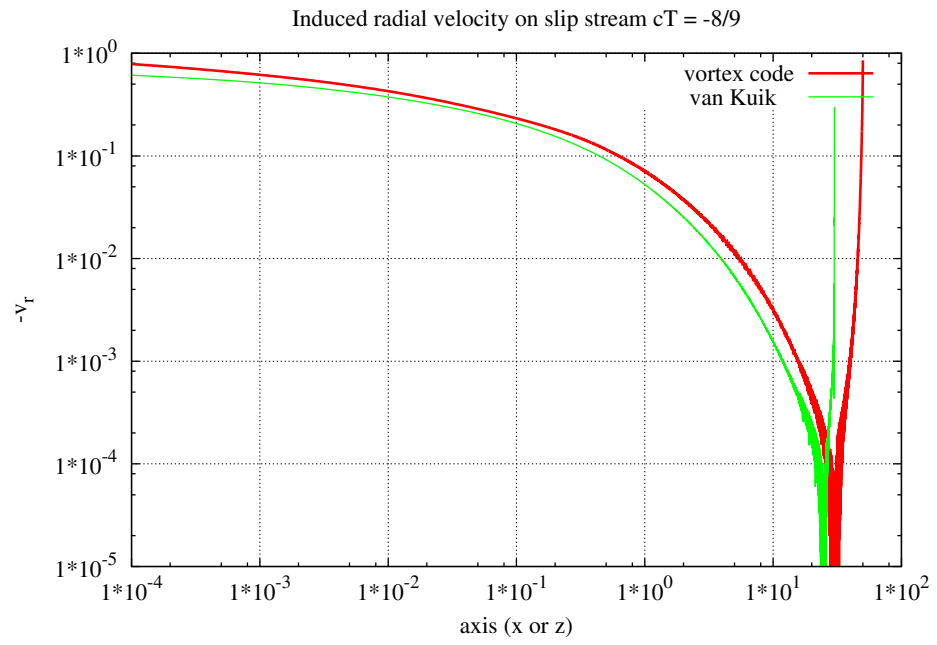


Figure 7: Calculated axial distribution of induced radial velocity, Betz case

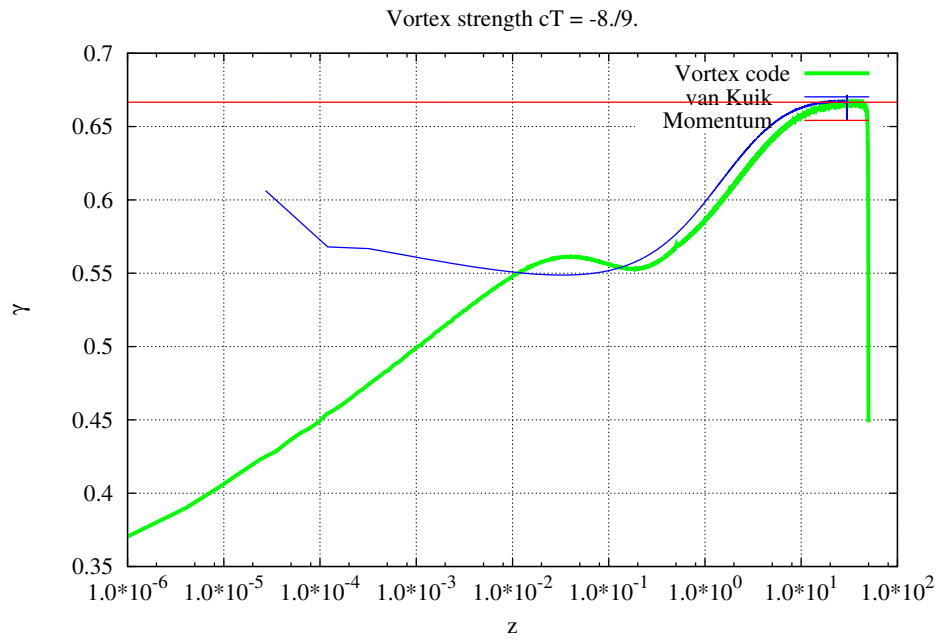


Figure 8: Axial distribution of vortex strength, Betz case.

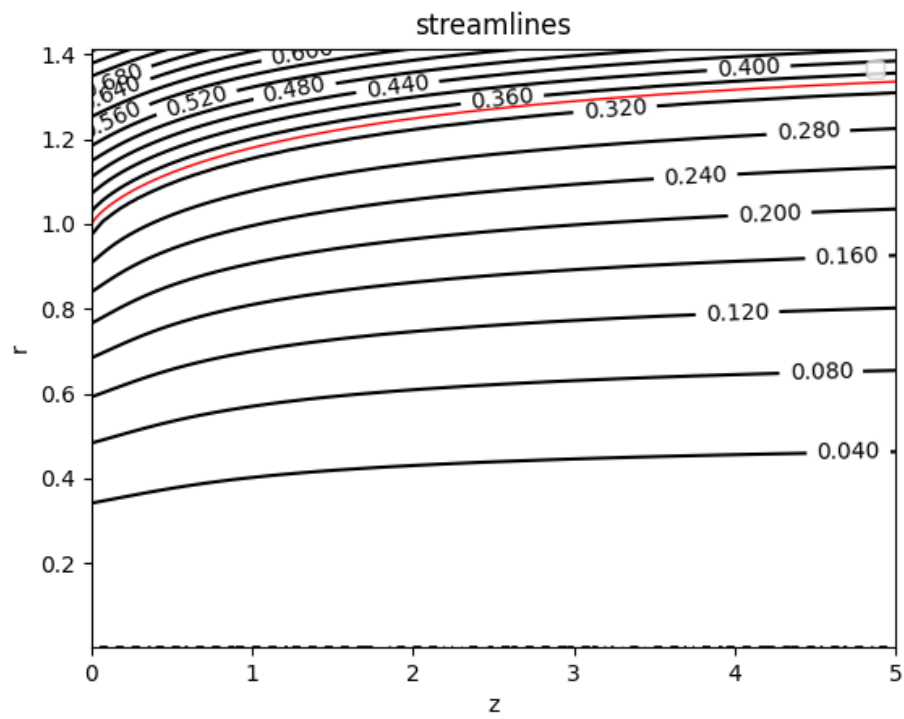


Figure 9: Stream-lines for the Betz case $c_T = -8/9$ red: slip stream, $\Psi_{wake} = 0.33333$

number of rings (k)	error in cP
4	$1.5 \cdot 10^{-4}$
8	$1.7 \cdot 10^{-5}$
16	$1.6 \cdot 10^{-6}$

Table 3: error in c_P as function of number of rings

6 Summary and Discussion

This report is about an implementation of a vortex ring/cylinder model for investigating the constantly loaded axi-symmetric Actuator Disk in the spirit of Øye, van Kuik, Bontempo and Manna. We confirm that the following model parameters influence its accuracy:

1. Number of rings,
2. Location of semi-infinite vortex cylinder,
3. Number of integration points at the disk,
4. Model for self-induction,
5. Shape of slip-stream,
6. Number and distribution of vortex rings close to the disk edge.

We therefore confirm findings from Gijs van Kuik [17] that some inter-relationship between parameters (see table 1) of the model exists. Nevertheless, it seems that an increase of number of rings always leads to an increase of accuracy. To summarize:

1. Model accuracy: Slip-stream iteration was stopped when using a special combination of model parameters deviations for c_P from momentum theory were $< 10^{-6}$. Table 3 gives an example of accuracy reached (regarding c_P from momentum theory) as function of number of rings:
2. Edge singularity: Van Kuik [16, 17] and Chattot [7] speculated about a possible edge singularity. Our findings, see for example Fig 8, indicate different behavior to both. As van Kuik already mentions in [17], Chattot [7] assumes some case of a constant vortex strength, which

seems not to be the case. Nevertheless, Fig. 8 for $10^{-6} < z < 10^{-2}$ (four decades) shows a logarithmic **decrease** in vortex strength. Both v_z (Fig. 6) and v_r (Fig. 7) tend to finite values for $z \rightarrow 0^+$.

3. Consistency: Do we fulfill theorem 1? **it seems so, perhaps.**

7 Acknowledgments

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